TripNest AI: Database Schema

Generated from the SQLAlchemy models (`backend/app/models.py`) on 2 November 2026; migration `70951402a50a_initial_schema`. SQLite is used for local development and PostgreSQL in production; the same models and Alembic migration serve both.

Separation of data

Kind of data	Where it lives	Notes
Destination and attraction information	Tables `states`, `destinations`, `attractions`	Seeded from `app/data/catalog.py` (public knowledge; indicative hours and fees; **assumed** carrying capacities)
Crowd observations (history)	**Not stored in the database.** Generated on demand by `services/crowd_sim.py` (demonstration data)	A real loader returning the same columns (`key, date, hour, count, rain`) replaces it
Model and predictions	Trained model file `backend/models/crowd_model.joblib` and `metrics.json`; predictions are computed per request and cached in memory	Keeps predictions reproducible from the model version

Tables used by the product

`states`

Column	Type	Notes
`id`	INTEGER	primary key
`code`	VARCHAR(4)	unique, required
`name`	VARCHAR(80)	unique, required
`kind`	VARCHAR(10)	required
`region`	VARCHAR(20)	required

`destinations`

Column	Type	Notes
`id`	INTEGER	primary key
`slug`	VARCHAR(80)	unique, indexed, required
`name`	VARCHAR(120)	required
`state_id`	INTEGER	FK to `states.id`, required
`lat`	FLOAT	required
`lon`	FLOAT	required
`categories`	JSON	required
`budget`	VARCHAR(10)	required
`daily_cost`	INTEGER	required

Column	Type	Notes
`min_days`	INTEGER	required
`best_months`	JSON	required
`climate`	VARCHAR(20)	required
`description`	TEXT	required
`how_to_reach`	TEXT	required
`facilities`	JSON	required
`featured`	BOOLEAN	required
`popularity`	FLOAT	required
`source`	VARCHAR(40)	required
`image_url`	VARCHAR(500)	nullable; Wikimedia photo URL (no file is stored)
`image_title`	VARCHAR(200)	nullable
`image_credit`	VARCHAR(300)	nullable; photographer or author
`image_licence`	VARCHAR(80)	nullable
`image_source_page`	VARCHAR(500)	nullable; Commons page
`updated_at`	DATETIME	required

`attractions`

Column	Type	Notes
`id`	INTEGER	primary key
`slug`	VARCHAR(100)	unique, indexed, required
`destination_id`	INTEGER	FK to `destinations.id`, required
`name`	VARCHAR(140)	required
`category`	VARCHAR(30)	required
`lat`	FLOAT	required
`lon`	FLOAT	required
`capacity`	INTEGER	required
`popularity`	FLOAT	required
`open_hour`	INTEGER	required
`close_hour`	INTEGER	required
`fee`	INTEGER	required
`sensitive`	BOOLEAN	required
`family_friendly`	BOOLEAN	required
`elder_friendly`	BOOLEAN	required
`covered`	BOOLEAN	required
`visit_min`	INTEGER	required
`tags`	JSON	required
`description`	TEXT	required
`source`	VARCHAR(40)	required

Relationships: a state has many destinations; a destination has many attractions (deleting a destination deletes its attractions).

Dormant tables (created, not used by the two core features)

These tables belong to modules that were unplugged in the scope change. They are kept so the modules can be restored; no rows are read by the product and none were dropped.

Table	Belonged to
`users`	accounts
`favourites`	accounts
`itineraries`	itinerary planner
`reviews`	reviews
`crowd_reports`	traveller reports
`listings`	local listings
`posts`	community
`comments`	community
`content_reports`	community moderation
`data_sources`	admin console

Current development-database contents

Table	Rows
`states`	36
`destinations`	84
`attractions`	271

Dormant tables hold only seeded demo rows (for example demonstration reviews); the product does not display them.

The `image_*` columns were added by Alembic migration `8a2f26c2f8b6_destination_photos`; for an older local SQLite file the app adds them at start-up (`db.add_missing_columns`).